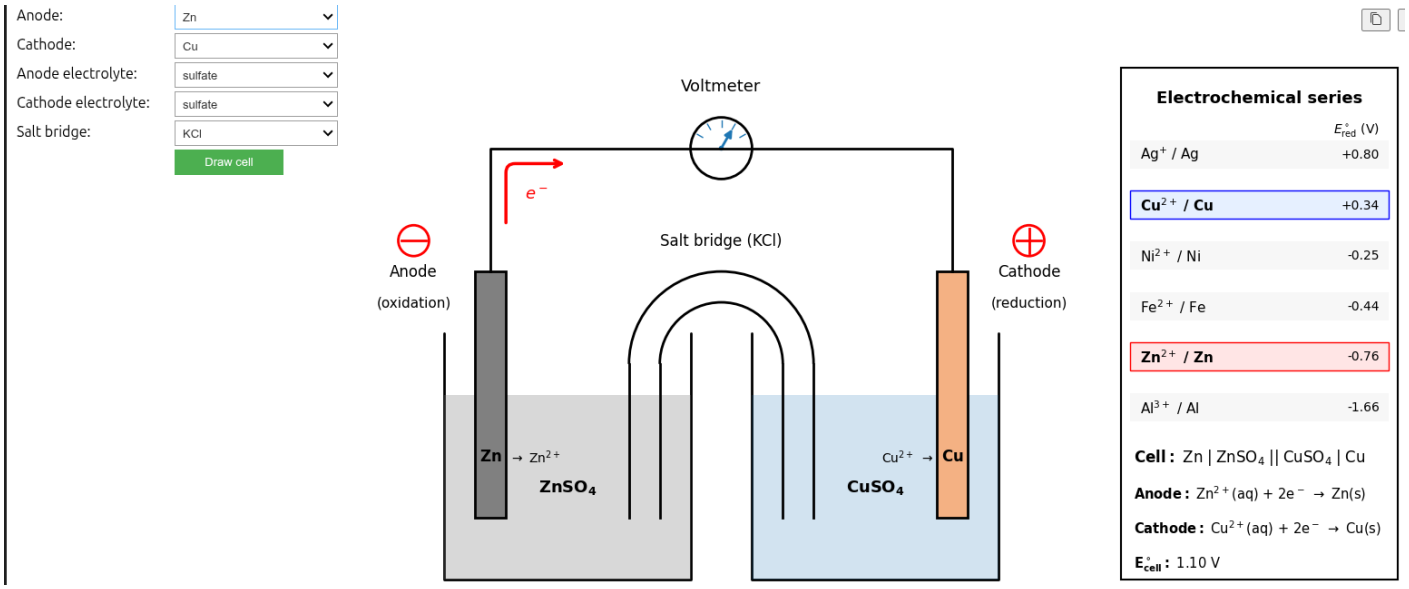

EXERCISE 1

Thermodynamics of electrochemical cells

April 16, 2026 | O. Schmidt | Theory and Computation of Energy Materials (IET-3)

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AIM OF TODAY'S EXERCISES

Today, we investigate the thermodynamic effects of electrochemical cells. The following link may be helpful

$\longrightarrow$ Exercises $\longleftarrow$

Objectives: The student is expected to be able to devise/ derive and/ or calculate

- The cell designation, the half cell reaction, the cell reaction, and the standard cell potential.
- The cell potential at non-standard conditions.
- The electrode potential.

when provided with sufficient material.

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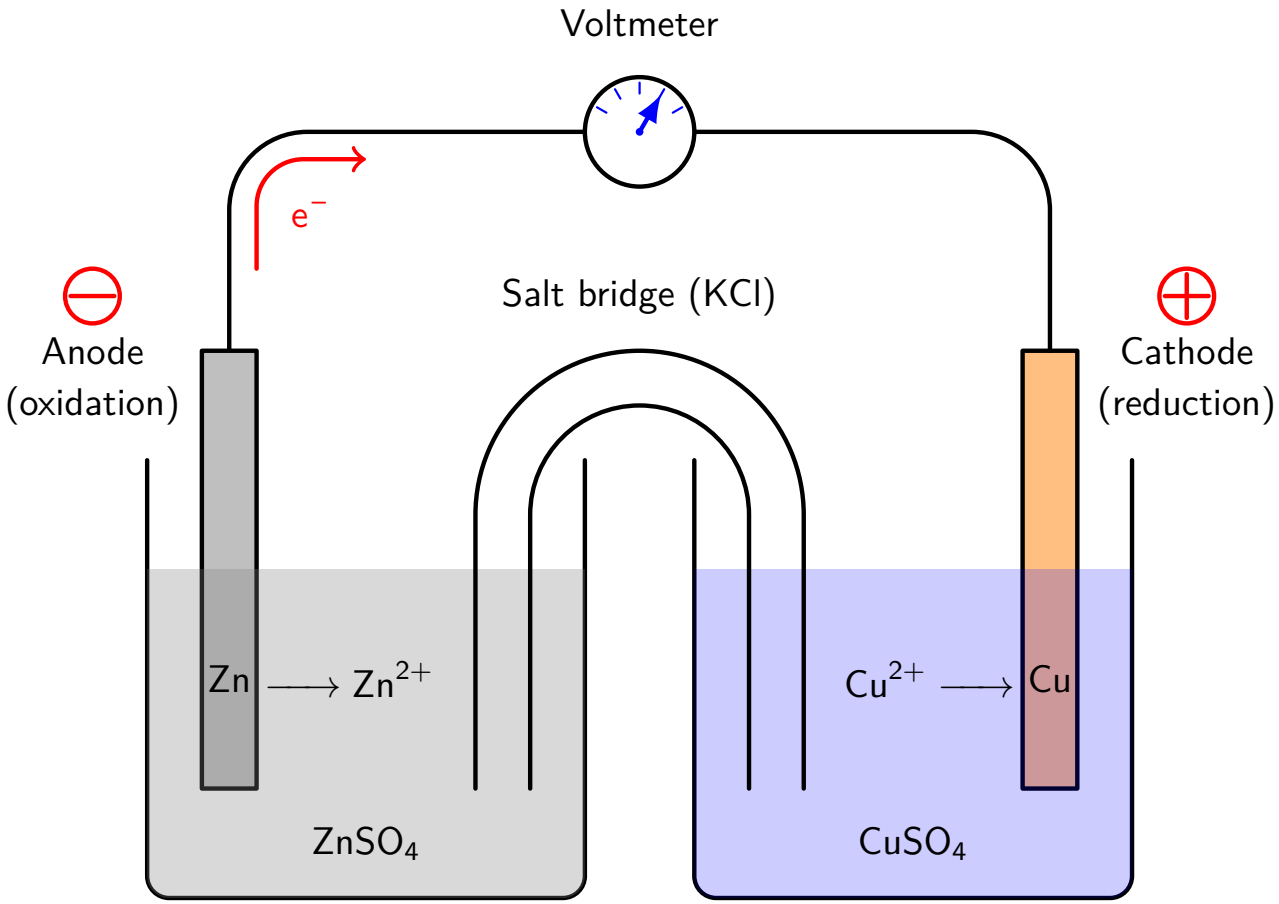
THE GALVANIC CELL: DANIELL CELL



THE GALVANIC CELL: WORKED EXAMPLE

Cell designation: We have the following cell desingation

$$\mathrm{Zn(s)|ZnSO_4(aq)||CuSO_4(aq)|Cu(s)}$$

Half-cell reactions: The half-cell reactions are

$$\begin{aligned} \text{Anode}: \quad & \mathrm{Zn^{2+}\,(aq) + 2\,e^- \longrightarrow Zn\,(s)} \\ \text{Cathode}: \quad & \mathrm{Cu^{2+}\,(aq) + 2\,e^- \longrightarrow Cu\,(s)} \end{aligned}$$

Cell reaction: The cell reaction is found to be

$$\mathrm{Cu^{2+}\,(aq) + Zn\,(s) \longrightarrow Cu\,(s) + Zn^{2+}\,(aq)}$$

Standard cell potential: Using the Table on slide 5

$$E^{\ominus}_{\mathrm{cell}} = E^{\ominus}_{\mathrm{red}}\left(\mathrm{Zn^{2+}/Zn}\right) - E^{\ominus}_{\mathrm{red}}\left(\mathrm{Cu^{2+}/Cu}\right) = 0.34\,\mathrm{V} - (-0.76\,\mathrm{V}) = \underline{\underline{1.10\,\mathrm{V}}}$$

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THE GALVANIC CELL: QUESTIONS

1.1) For the following half cell reactions

$$2\,\mathrm{Ag}^{+}\,(\mathrm{aq}) + \mathrm{Ni}\,(\mathrm{s}) \longrightarrow 2\,\mathrm{Ag}\,(\mathrm{s}) + \mathrm{Ni}^{2+}\,(\mathrm{aq})$$

- 1.1a) Write the half-cell reactions.
- 1.1b) Calculate the standard cell potential.

1.2) Draw a diagram of the following galvanic cell with a salt bridge of your choice

$$\mathrm{Al}\,(\mathrm{s})\,|\mathrm{Al}\,(\mathrm{NO_3})_3\,(\mathrm{aq})\,||\mathrm{Fe}^{2+}\,(\mathrm{aq})\,|\mathrm{Fe}\,(\mathrm{s})$$

1.3) On slide 6, there is a diagram. For this diagram, please

- 1.3a) Write the cell designation.
- 1.3b) Write the half-cell and cell reaction(-s).
- 1.3c) Calculate the standard cell potential.

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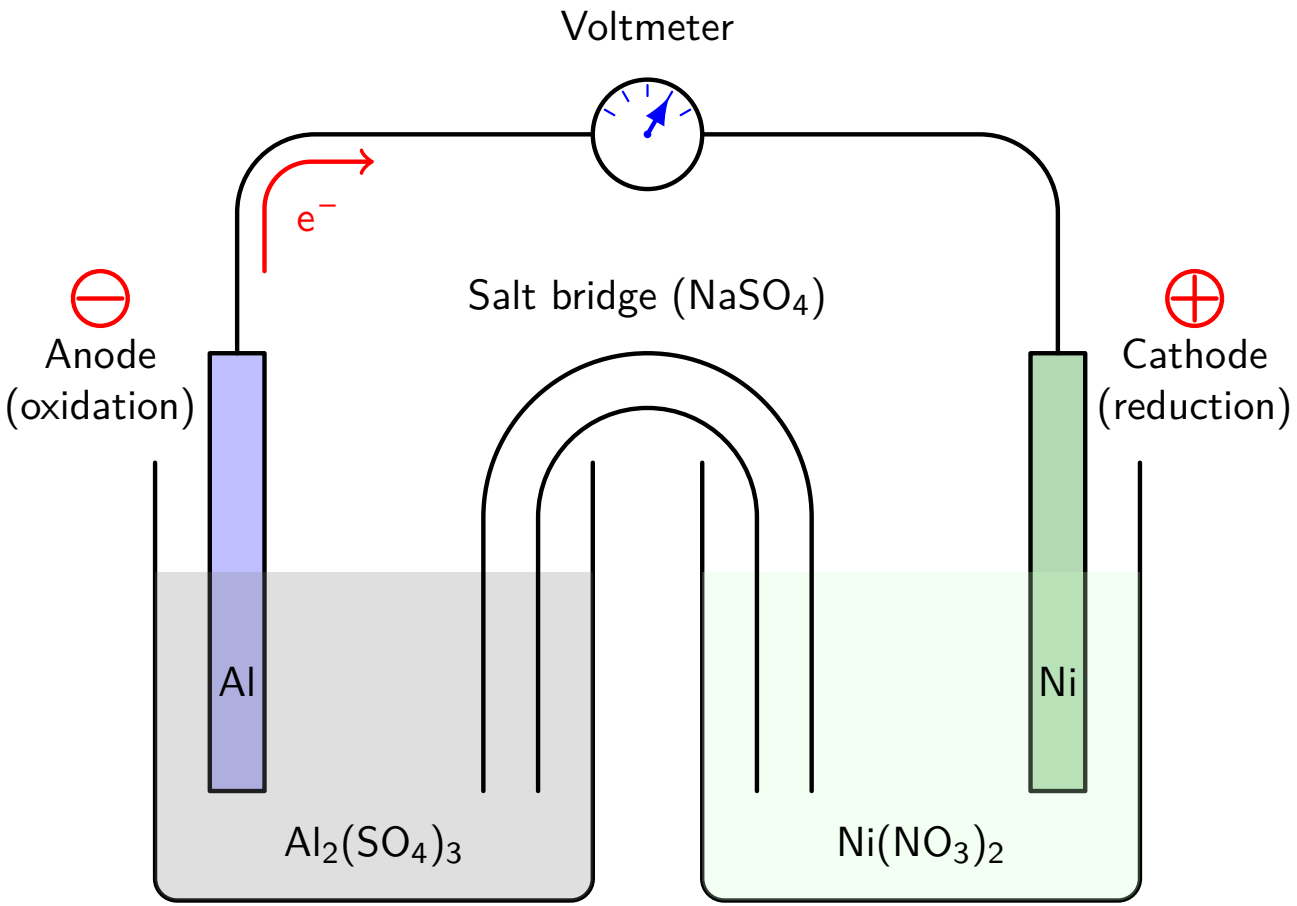
ELECTROCHEMICAL SERIES

Half-cell			$E^{\ominus}_{\mathrm{red}}$ **(V)**
$\mathrm{Ag^+(aq) + e^-}$	$\rightarrow$	$\mathrm{Ag(s)}$	$+0.80$
$\mathrm{Cu^{2+}(aq) + 2e^-}$	$\rightarrow$	$\mathrm{Cu(s)}$	$+0.34$
$\mathrm{Ni^{2+}(aq) + 2e^-}$	$\rightarrow$	$\mathrm{Ni(s)}$	-0.25
$\mathrm{Fe^{2+}(aq) + 2e^-}$	$\rightarrow$	$\mathrm{Fe(s)}$	-0.44
$\mathrm{Zn^{2+}(aq) + 2e^-}$	$\rightarrow$	$\mathrm{Zn(s)}$	-0.76
$\mathrm{Al^{3+}(aq) + 3e^-}$	$\rightarrow$	$\mathrm{Al(s)}$	-1.66

* Values are measured at 1 bar and 298.15 K.

THE GALVANIC CELL: QUESTION 1.3



THE CELL POTENTIAL

Nernst equation

Let E_{cell} be the cell potential and Q be the reaction quotient of an electrochemical reaction, then the cell potential is found using the Nernst equation:

$$\textbf{Nernst}' \textbf{ equations}: \qquad E_{\mathrm{cell}} = E_{\mathrm{cell}}^{\ominus} - \frac{RT}{\nu F} \ln(Q)$$

Furthermore, at equilibrium we have $Q = K$, and no potential is generated by the galvanic cell, i. e. $E_{\mathrm{cell}} = 0$. Therefore

$$\textbf{Equilibrium}: \qquad E_{\mathrm{cell}}^{\ominus} = \frac{RT}{\nu F} \ln(K) \implies K = \exp\left(\frac{\nu F\, E_{\mathrm{cell}}^{\ominus}}{RT}\right)$$

THE CELL POTENTIAL: WORKED EXAMPLE

Consider the Daniell cell example from before.

Number of electrons, ν: We repeat the half-cell reactions for completion here and box the number of equal equation

$$\text{Anode}: \quad \mathrm{Zn^{2+}\,(aq)} + \boxed{2\,\mathrm{e^-}} \longrightarrow \mathrm{Zn\,(s)}$$

$$\text{Cathode}: \quad \mathrm{Cu^{2+}\,(aq)} + \boxed{2\,\mathrm{e^-}} \longrightarrow \mathrm{Cu\,(s)}$$

i. e. the number of electrons in the half-cell reactions is $\nu = \underline{\underline{2}}$.
The equilibrium constant: Insert the values

$$K = \exp\left(\frac{\nu F\, E^{\ominus}_{\mathrm{cell}}}{RT}\right) = \exp\left(\frac{2 \times 96485\,\mathrm{C\,mol^{-1}} \times 1.10\,\mathrm{V}}{8.314\,\mathrm{J\,mol^{-1}} \times 298.15\,\mathrm{K}}\right) \approx \underline{\underline{1.55 \times 10^{37}}}$$

THE CELL POTENTIAL: QUESTIONS

2.1) For the following half cell reactions

$$2\,\mathrm{Ag}^{+}\,(\mathrm{aq}) + \mathrm{Ni}\,(\mathrm{s}) \longrightarrow 2\,\mathrm{Ag}\,(\mathrm{s}) + \mathrm{Ni}^{2+}\,(\mathrm{aq})$$

2.1a) How many electrons participate in the half-cell reactions?

2.1b) Calculate the equilibrium constant of the reaction under standard conditions.

2.2) Consider the following cell

$$\mathrm{Fe(s)|FeSO_4(aq)||NiCrO_4(aq)|Ni(s)}$$

how much does the cell potential change, when Q is increased by a factor 10?

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CHEMICAL POTENTIAL IN THE IDEAL DILUTE LIMIT

Activity: Chemical potential in the dilute limit

Let $\mu_{\mathrm{A}}^{(\mathrm{pure})}$ be the chemical potential of a mixture of pure A, a_{A} be the activity of compound A, μ_{A} be the chemical potential of compound A. Then the chemical potential is provided by

$$\mu_{\mathrm{A}} = \mu_{\mathrm{A}}^{(\mathrm{pure})} + RT \ln\left(a_{\mathrm{A}}\right)$$

and in the ideal-dilute limit, we have

$$\mu_{\mathrm{A}} \approx \mu_{\mathrm{A}}^{(\mathrm{pure})} + RT \ln\left(x_{\mathrm{A}}\right)$$

which is shown in the appendix.

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APPENDIX: IDEAL LIMITS

Ideal case: Raoult's law

Consider an arbitrary compound A. Let $p_{\mathrm{A}}^{(\mathrm{pure})}$ be the vapour pressure of pure A, $p_{\mathrm{A}}^{(\mathrm{solution})}$ be the vapour pressure of A immersed in a solution, and x_{A} be the mole fraction. For an ideal solution, the following holds true

$$p_{\mathrm{A}}^{(\mathrm{solution})} = p_{\mathrm{A}}^{(\mathrm{pure})} x_{\mathrm{A}}$$

Ideal-dilute case: Henry's law

Let K_{A} be an empirical constant, then the following holds true in for an ideal dilute solution

$$p_{\mathrm{A}}^{(\mathrm{solution})} = K_{\mathrm{A}} x_{\mathrm{A}}$$

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APPENDIX: ACTIVITY

Activity and ideal solution

Consider a solvent with compound A. The activity of compound A is defined by

$$\mu_{\mathrm{A}} = \mu_{\mathrm{A}}^{(\mathrm{pure})} + RT \ln\left(a_{\mathrm{A}}\right)$$

and we can measure this value using the vapour pressure

$$a_{\mathrm{A}} = \frac{p_{\mathrm{A}}^{(\mathrm{solution})}}{p_{\mathrm{A}}^{(\mathrm{pure})}}$$

APPENDIX: ACTIVITY + IDEAL DILUTE

Activity: Chemical potential in the dilute limit

Applying the activity definition, measurement expression, Henry's law and $\log(a\,b) = \log(a) + \log(b)$

$$\mu_{\mathrm{A}} = \mu_{\mathrm{A}}^{(\mathrm{pure})} + RT\ln(a_{\mathrm{A}}) = \mu_{\mathrm{A}}^{(\mathrm{pure})} + RT\ln\left(\frac{p_{\mathrm{A}}^{(\mathrm{solution})}}{p_{\mathrm{A}}^{(\mathrm{pure})}}\right)$$

$$= \mu_{\mathrm{A}}^{(\mathrm{pure})} + RT\ln\left(\frac{K_{\mathrm{A}}x_{\mathrm{A}}}{p_{\mathrm{A}}^{(\mathrm{pure})}}\right) = \mu_{\mathrm{A}}^{(\mathrm{pure})} + RT\ln\left(\frac{K_{\mathrm{A}}}{p_{\mathrm{A}}^{(\mathrm{pure})}}\right) + RT\ln(x_{\mathrm{A}})$$

And if Raoult's law holds, then $x_{\mathrm{A}}p_{\mathrm{A}}^{(\mathrm{pure})} = K_{\mathrm{A}}x_{\mathrm{A}} \implies p_{\mathrm{A}}^{(\mathrm{pure})} = K_{\mathrm{A}}$ and from $\log(1) = 0$, we have

$$\mu_{\mathrm{A}} = \mu_{\mathrm{A}}^{(\mathrm{pure})} + RT\ln(x_{\mathrm{A}})$$

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